

Ref Books:① TCP/IP Protocol Suite (4th Edition)

- Forouzan

② Computer networking Top Down Approach (8th)

- Ross, Kurose

Lecture-2

sunday

10/8/25

Chapter-2

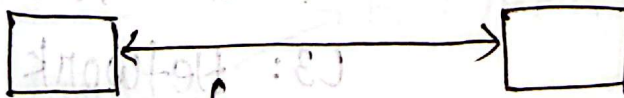
↳ [Forouzan]

OSI Model & TCP/IP suite⇒ OSI:

→ OSI model একটা Protocol. যার দ্বারা device ও Protocol ভিত্তি করে, অর্থাৎ n/w এর সাথে connect হবে।

⇒ TCP/IP Protocol suite:

↳ Another set of Protocol



↓
কিভাবে commⁿ হবে, সেখানে OSI / TCP/IP Protocol.

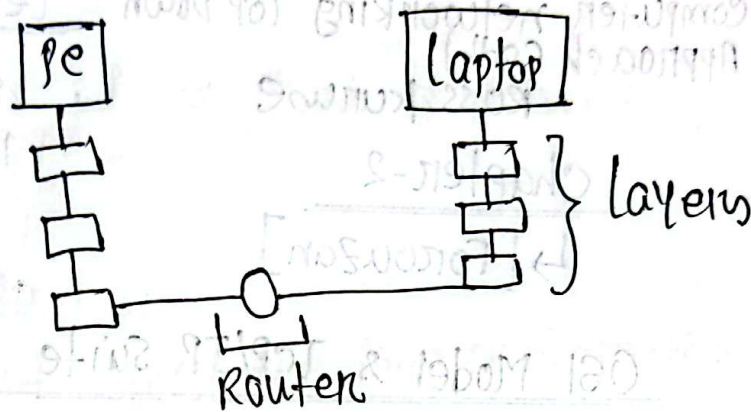
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→ দুইটা device, এর comm" এর জন্য, আরো অনেক layer থাকে।



* OSI Model কয়টা middlewear.

OSI - Model Layers: TCP/IP Model layers:

7 layers -

5 layers

L7 : Application

L6 : Presentation

L5 : Session

L4 : Transport

L3 : Network

L2 : Data link

L1 : Physical

L5 : Application

Presentation

Session

L4 : Transport

L3 : Network

L2 : Data link

L1 : physical.

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- * Send data layer top to Bottom.
- * Receive " " Bottom to top.
- * Every layer at most has adjacent layer and comm'n with it.
- * Every layer data and extra header + trailer add.

(***) OSI vs TCP/IP

	OSI	TCP/IP
Innovation	New	Old
Used	less	Much

⇒ Main of TCP/IP model

Key Terms:

① IP Address → H/W layer (L3)

② Physical Address (MAC) → Data link layer (L2)

③ Port number/Address → Transport layer (L4)

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TOPIC NAME :

layer

Data vs other

DAY :

TIME :

DATE :

* Application → message

* Transport → segment

* Network → datagram

* Data link → frame

* physical → bits.

Sadi Sir

ChP-2

Lecture-3

TOPIC NAME: Application Layer

DAY: Wednesday

TIME:

DATE: 13 / 8 / 25

→ Rules & Regulation বিধি আন্দোলন করা হয়।

Principle
of H/W App

Web & HTTP

→ User ten internet use করতে হয়।

Application layer
↓
Transport layer

⊗ Email → SMTP

Web → HTTP

File → FTP (ফাইল ট্রান্সফার প্রোটোকল)

Creating a network app:

① Write program

② No need to write software for new come device

Application Architecture:

① client server

② peer to peer

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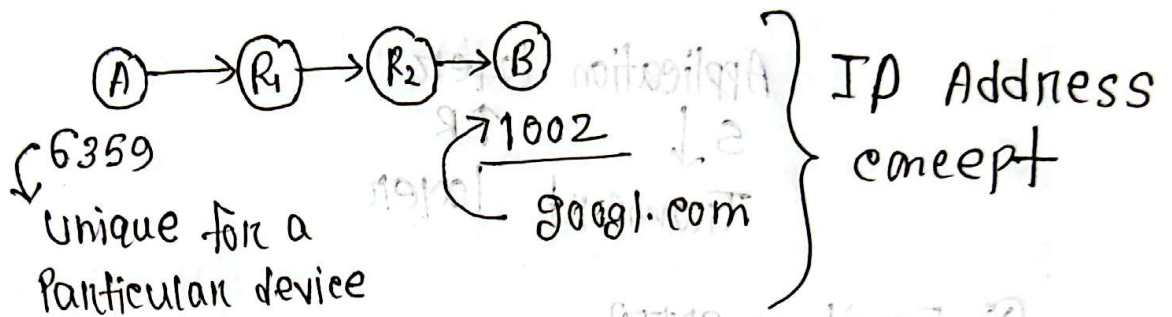
Client server

Server:

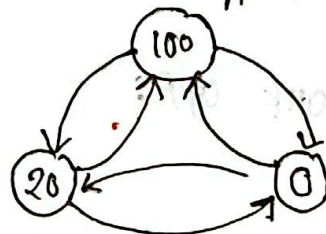
- ① Server always on
- ② Permanent IP Add

Client:

- ① client = end user.



Peer to peer



Process commⁿ

→ प्रक्रिया running application की प्रक्रिया.

Socket

Application के मध्य पारस्परिक संपर्क.

TOPIC NAME: _____

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Ports

→ ~~Port~~ ports have numerical values.

Addressing values

Client (1001)

YT (2452)



⇒ IP Address : Port Number

HyperText Transfer Protocol

→ অনেকটা client server architecture এর মতো।

Protocol : // Host : Port / Path

↓
Method (http, https)

⊗ HTTP → Port = 80

⊗ HTTPS → Port = 443

⊗ HTTP uses TCP

⊗ HTTP is stateless → কে রেজ করছে, তা মনে রাখে না।

Sadi Sir

Leetune-4

TOPIC NAME: chp-2 Application layer

DAY: Sunday

TIME:

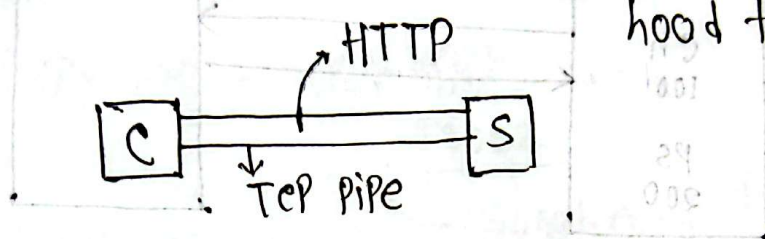
DATE: 17/8/25

Round Trip Time (RTT)

RTT = $C \rightarrow S$ Req karta tashq, Receive karta jab
↖
qayt time.

⊗ Reliable $\rightarrow$ TCP

⊗ HTTP under the hood tep me karta.

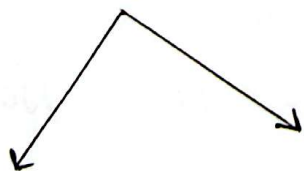


TCP 3-way Handshake

$C \xrightarrow{SYN} S \rightarrow C \rightarrow S$

HTTP connection

Non persistent



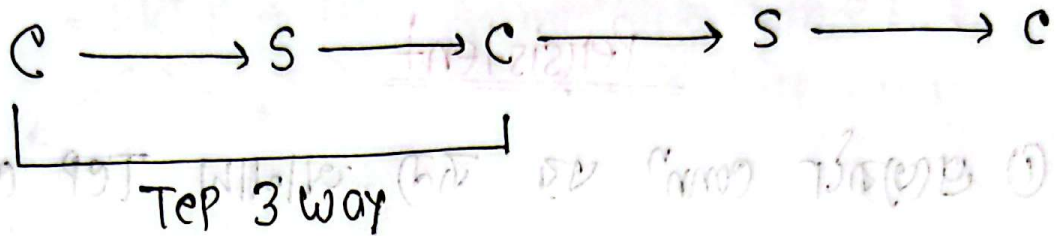
GOOD LUCK

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⇒ Without a parallel connection:



⊗ At a time only HTTP Req execution 20.

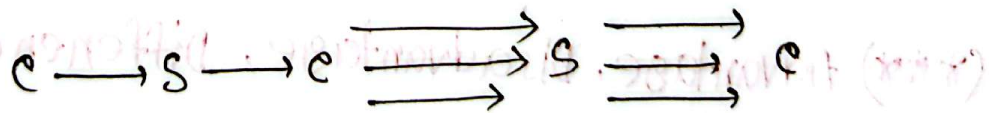
eg: 1 web page } 1st तर झुडू web page
20 image } load 20.

execution 23 तर तर execution close 23 तर 200.

⊗ Sequentially execution 23 तर.

⇒ A Parallel connection:

⊗ Parallely 23 तर 20.



⊗ Non Persistent : New connection + file transfer

Times = (11) * 21

TOPIC NAME: _____

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(***) Diagram এর explanation.

Persistent

① প্রত্যেকের conn এর জন্য আলাদা TCP connection open করা লাগে না। একবার connection establish করলে হয়।

* Persistent : 1 connection + (transfer) * 20

=> Non Pipeline:

Aek বাতয়ার পরে আরো Res পাঠানো।

=> Pipeline:

একবার সব send করে দিব।

(***) Advantage, Disadvantage, Difference এর
মধ্যে ব্যতীত।

* Conn establish করতে RTT পরিমাণ time লাগে।

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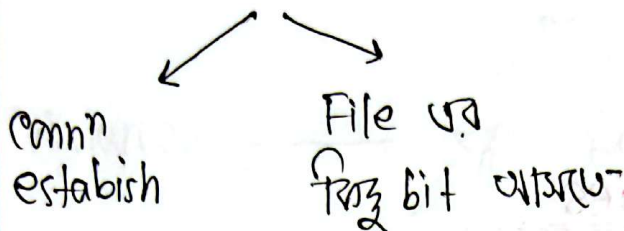
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⊛ Non Persistent HTTP Resume time

$$= 2RTT + \text{File transmission time}$$



1 wp + 10 img → (Non Persistent)

$$2RTT + (2RTT + \text{Tran}^m \text{ time}) * 10 = \text{Total time}$$

Persistent

$$\text{Total time} = 2RTT + (RTT + \text{Tran}^m \text{ time}) * 10$$

HTTP Protocol - Message Format

→ शुरुआत
जहाँ
मैसेज

C

S

GOOD LUCK

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

HTTP status code

201 : Status OK

404 : Not found

cookies

→ HTTP stateless, but cookie is used to maintain the state.

→ First party cookie } (***)
→ Third party cookie }

↳ 3rd party

cookie in the track of the user. (***)

Sadi Sir

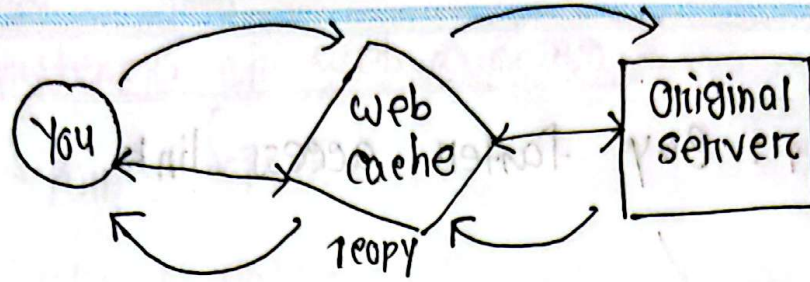
Lecture-5

TOPIC NAME: Web cache (Proxy server)

DAY: Sunday

TIME:

DATE: 24/8/25



- ⊗ Web cache এর কাজ, অর্থাৎ এর component থাকে না।
- ⊗ Hit miss রকম, Original server এর কাজ রেখে দেবে, then তার একটি copy নিজের কাজ রাখে থাকে।

Avg data rate to browsers = $\frac{\text{Web object size}}{\Delta} \times \text{Avg req rate}$

Access link utilization = $\frac{\Delta}{\text{access link rate}} = 0.97$

$\rho \approx 1 \rightarrow$ queuing delay (s)

$\rho < 0.6 \rightarrow$ milis

lan utilization = $\frac{\Delta}{\text{LAN speed}} = 0.0015 \rightarrow (\mu s)$

⊗ Utilization রকম = অল্প রকম থাকবে।

end-to-end Delay = internet delay(RTT) + access link delay + LAN delay
 $= 2s + \text{min} + \mu \text{sec}$

TOPIC NAME: _____

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Soln:

⇒ Option 1: Buy faster access link

↙ costly

⇒ Option 2: Install a web cache.

↙ cache hit = 40% = 0.4

cache miss = 60% = 0.6

Rate to browser over access link = 0.6×1.5
= 0.9

Access link utilization = $\frac{0.9}{1.54} = 0.58$ (m sec)

Avg end-to-end delay = $0.6 \times (\text{delay from original server} + \text{queuing}) + 0.4 \times \text{delay when cache hit}$

= $0.6 (2s + 0.01s) + 0.4 (ms)$

↓
mili s 3 मेगा 2000

= $\sim 1.2s$

↙ easier + less costly

TOPIC NAME: _____

DAY: _____

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Consistency of web caching

↓ pull ↓ push

Conditional Rate:

↓ HTTP status code = 304

Content Delivery Network (CDN)

↓ self study (must read also)

HTTP/2 vs HTTP 1.1 (***)

↓ HoI (Head of the line) Blocking → Must read also !

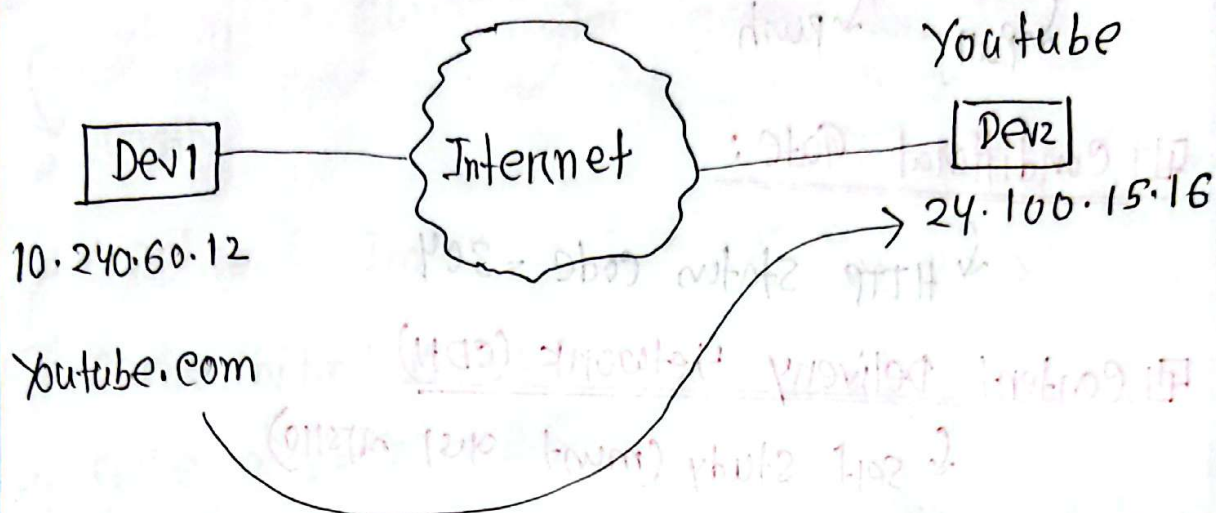
solⁿ { HTTP/2 → Req & Res ko alag frame v chl krta
(***) shren, Multiplexing kr krkr krta. Req
Order ki change krta krta,
(concept important (***))

TOPIC NAME : DNSDAY: Sunday

TIME:

DATE: 31/8/25

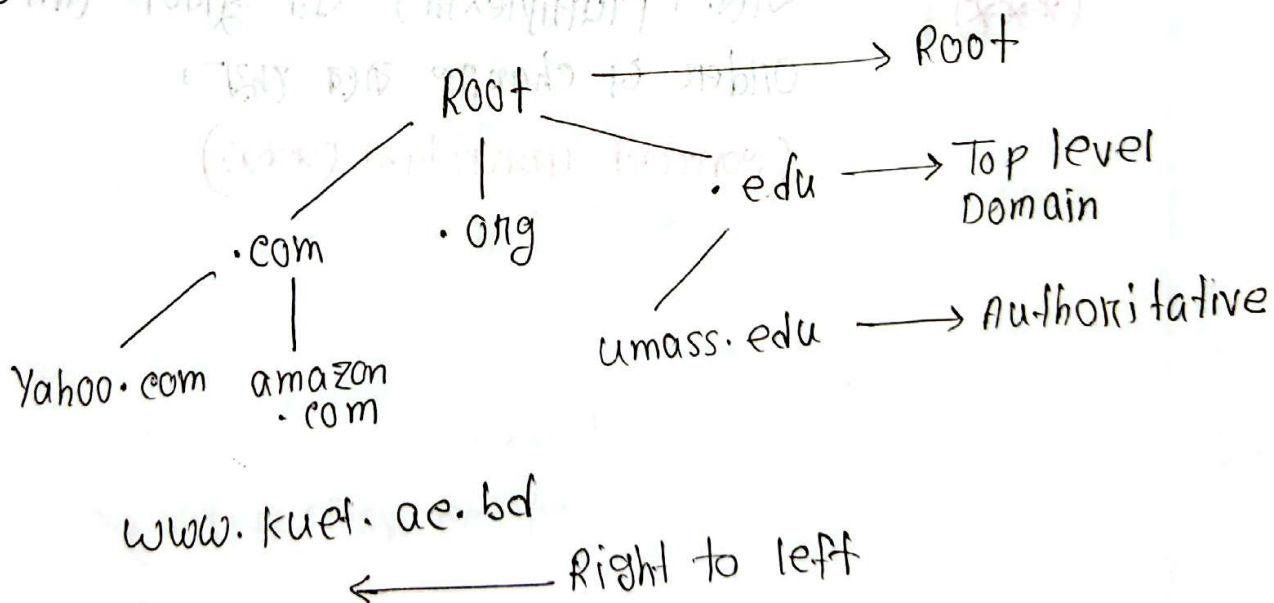
(*) Application layer ka kora CT.



*) IP Address ka kora kare Abstraction (karta hai) Number $\rightarrow$ String & map karta hai DNS.

*) DNS kare application layer service.

*) DNS - hai centralized hai decentralized?



TOPIC NAME : _____

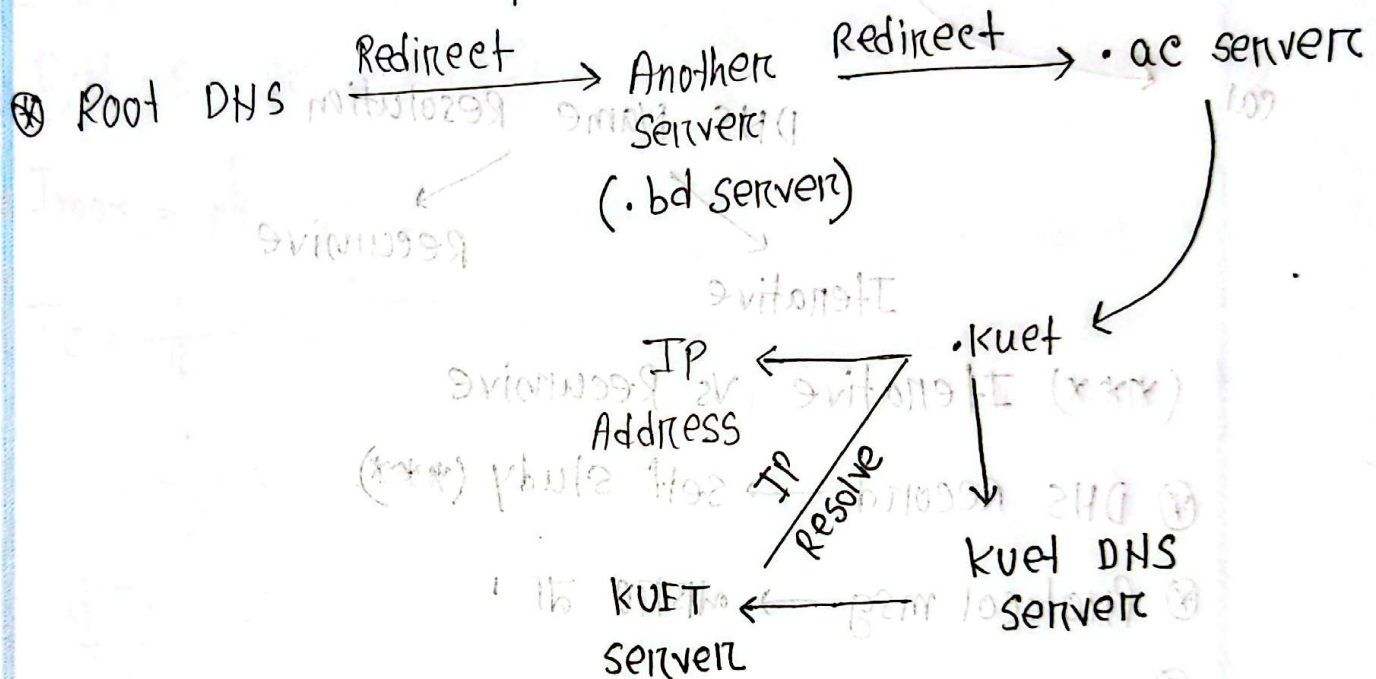
DAY : _____

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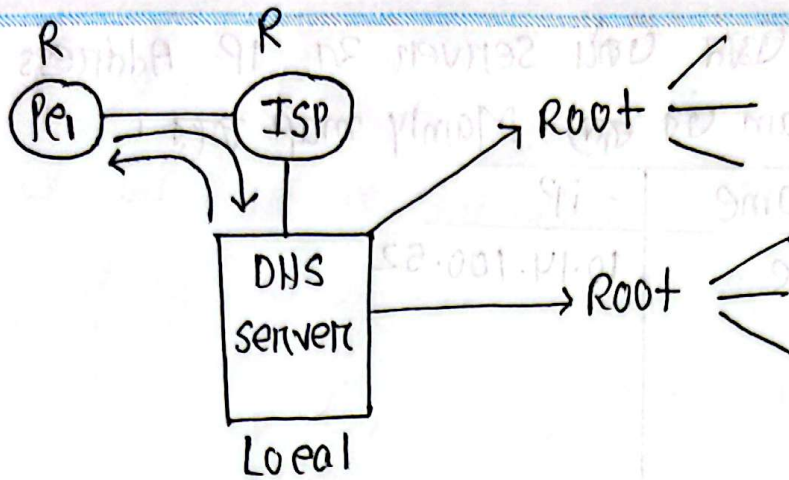
⇒ DNS server ডান একে server ২১ IP Address provide করে, একে Domain এর জন্য। Mainly map করে ...

name	ip
Youtube	10.14.100.52



⊗ Explain why DNS is Decentralized (***)

↓ প্রতিটি server এর আলাদা কপি থাকে। তাই কোনো একে server down থাকলে, অন্য server এ hit করা যায়।



DNS Name Resolution

↙ ↘

Static Recursive

(***) Iterative vs Recursive

⊗ DHS Record → self study (***)

④ Protocol msg $\rightarrow$ भाग 70 तः ।

Sadi Sir

TOPIC NAME: Email Ed-4

Top IP protocol suite (ehp-Email)

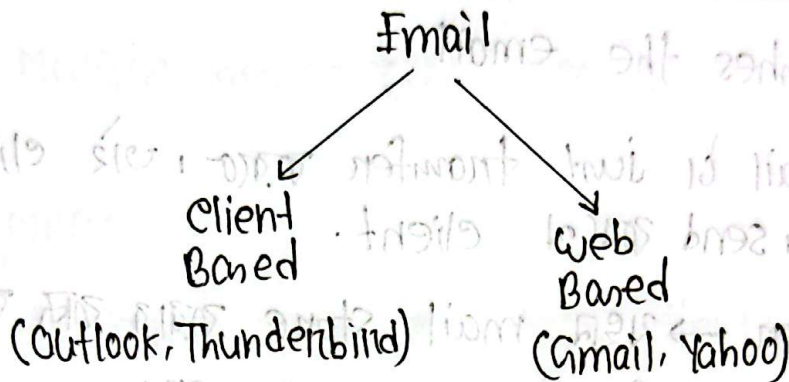
Lecture-7

DAY: wednesday

TIME:

DATE: 3 / 9 / 25

- ① User Agent
 - Outlook
 - Email client
- ② MTA
- ③ MAA



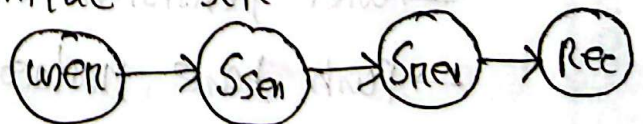
Mail Email component

- ① User agent
- ② Mail server → Email. स्थानांतरण स्थान
- ③ Protocol (SMTP) →

```
graph LR; S --> E; E --> S;
```

Mail server:

- ① Mail Box → अलग जग unique ser ser
- ② Mail queue →
- ③ SMTP
 - ↳ Architecture.



TOPIC NAME : _____

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Email Architecture :

Senden, Receive same mail server

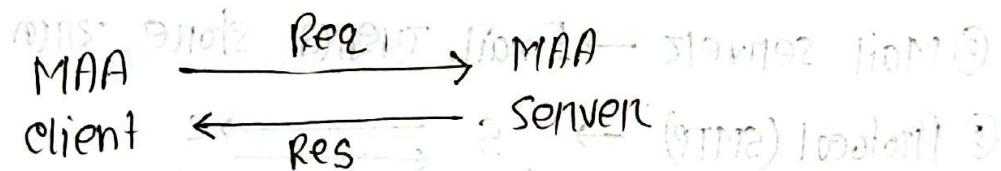
⊗ SMTP → Senden ২য় Rec ও Server এ থাকে।

→ Just pushes the email

⊗ MTA → Email ৬i just transfer করে। তবে client হিসাবে কাজ করে। send করেন client.

⊗ MTA server → যখন mail store করার কাজ করে।
Receive করেন server।

⊗ MAA → Message Access Agent



ও server
⊗ Receiver - যখন ওর machine এ থাকে we use POP3, IMAP.

⊗ SMTP :

→ User persistent connⁿ

→ Push base protocol

→ Can send 7bit ASCII code

→ SMTP + MIME

TOPIC NAME : _____

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POP: Post Office Protocol

- Rx ও server থেকে mail নিচে, client কে পাঠানো।
- Then mail server থেকে delete করে দিবে।
- Multiple device থেকে access করা যায় না।

IMAP:

- Server থেকে copy করে client এ পাঠানো।
- Then client কে পাঠানো।
- Multiple device থেকে access করা যায়।
- Storage Problem

Web Based Email: (Self Study) → HTTP কীভাবে connetate করে উঠে।

- Plain + HTTP transaction
- HTTP server
- যেহেতু HTTP দিয়ে হচ্ছে, jml নাম টা Replace
করে দিচ্ছি।
- HTTP Abstraction, underline SMTP-ই কাজ করে।

⊗ MIME jml convert করা, যাও SMTP use করেই ASCII
use করা যায়।

⊗ state বিনা বাহ্যে SMTP.

GOOD LUCK